

Mathematical Foundations of Generative Models

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

Date: 2026-10-06. Baseline: October 6, 2026.

HONESTY NOTE: unit order is the builder's provisional bridge order, not asserted official syllabus. See README.md and source_gaps.md G1-G2.

The spine

Generative modeling asks one question: how do we learn a probability law from samples, and then draw new samples from what we learned? The eight units answer it in order of mathematical dependency.

Unit dependency graph

U01 Probability, measures, and density estimation (P01, P06, P07, P08) -> U02 Latent models and variational inference (P05, P08, P18) -> U03 Autoencoders and representation objectives (P04, P11, P18) -> U04 Change of variables and normalizing flows (P04, P05, P08) -> U05 Adversarial and ratio-based estimation (P08, P09, P11) -> U06 Scores, diffusion, and stochastic dynamics (P05, P06, P18) -> U07 Autoregressive models and modern bridges (P13, P14, P15) -> U08 Evaluation, inference, and experimental synthesis (P07, P10, P22)

Cross links: U01 feeds every unit. U02 feeds U03 (VAE), U05 (divergence choice), and U06 (variational view of diffusion). U04 and U07 both give exact likelihood and are compared in U07-C12 and U08. U05 and U06 meet in score-based versus adversarial tradeoffs in U08. U08 closes the loop back to U01: held-out likelihood is density estimation judged honestly.

Unit summaries

- U01: the objects. Probability versus density, support, normalization,

discrete and continuous variables, joint and marginal laws, sampling, likelihood, the empirical distribution, estimation, entropy, divergence, identifiability. File: lessons/u01.md.

- U02: the intractable integral. Latent marginalization, posterior,

Bayes, Jensen, the ELBO identity, the variational gap, forward versus reverse KL, reparameterization, Monte Carlo variance, amortization, expressive families, degeneracy. File: lessons/u02.md.

- U03: codes that reconstruct and codes that generate. Deterministic

reconstruction, generative versus representation goals, VAE, Gaussian KL in closed form, posterior collapse, beta weighting, discrete latents, codebooks, straight-through bias, likelihood choice, rate/distortion, evaluation. File: `lessons/u03.md`.

- U04: exact likelihood by construction. Invertibility, determinant as

volume, Jacobian, the density transform, coupling and autoregressive flows, triangular Jacobians, base densities, exact likelihood, sampling direction, computational asymmetry, expressivity, numerical constraints. File: `lessons/u04.md`.

- U05: learning without a density formula. Implicit distributions,

density ratios, the discriminator optimum, the GAN objective, divergence assumptions, optimal transport, the Lipschitz constraint, WGAN and gradient penalty, mode collapse, identifiability, stability, evaluation limits. File: `lessons/u05.md`.

- U06: gradients of log-density. Score, score matching, denoising,

the noising chain, reverse kernels, the DDPM loss, DDIM, Langevin dynamics, SDE/ODE assumptions, conditioning and guidance, solvers, support and boundary behavior. File: `lessons/u06.md`.

- U07: the chain rule at scale. Chain rule factorization, conditional

densities, discrete tokens, teacher forcing, causal networks, the transformer as density model, sampling, exact likelihood, context cost, masked objectives, flow matching as a labeled extension, model comparison. File: `lessons/u07.md`.

- U08: judging generators honestly. Held-out likelihood, bound versus

exact scores, mode coverage, sample precision and recall, FID limits, memorization, compute-matched baselines, synthetic ground truth, uncertainty, falsification, the source-identity report, the role gap map. File: `lessons/u08.md`.

Files per unit

Each unit ships: `lessons/u0N.md`, `lessons/keys/keys-u0N.md`, `labs/lab-u0N.md`, `labs/keys/keys-u0N.md`, `visuals/u0N/*.png` with `visuals/render_u0N.py`. Shared banks: `interview/transfer-sets.md`, `interview/keys-transfer.md`, `interview/oral-defenses.md`, `interview/keys-oral.md`. Capstones: `capstones/capstone-research-gp-critic.md`, `capstones/capstone-applied-fde.md`.

Diagnostics

prerequisites.md holds the entry diagnostic: 12 questions, one per bridge area, with a pass bar and a remediation pointer per question. Take it before U01. A fail on any bridge sends the learner to the shared P-module and the unit-local remediation, not forward.

Prerequisites

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Shared modules live in `workspace/stanford-frontier-ai/v2-pack/shared/prerequisites/`. Each entry below links a shared module and names the unit-local remediation that teaches the same idea inside the lesson. A link is navigation. The local remediation is the teaching.

Bridge map

Code	Module	File	Used by	Local remediation
P01	Numeracy, algebra, notation	p01_numeracy.md	U01	U01 toy arithmetic before symbols
P04	Spectral and numerical linear algebra	p04_spectral.md	U03, U04	U03-C04, U04-C02 local derivations
P05	Scalar and multi-variable calculus	p05_calculus.md	U02, U04, U06	U02-C04, U04-C03, U06-C01 local derivations
P06	Probability, events to distributions	p06_probability.md	U01, U06	U01 is the local rebuild
P07	Statistical estimation and uncertainty	p07_estimation.md	U01, U08	U01-C09, U08-C01 local derivations
P08	Information theory and density objectives	p08_information.md	U01-U06	U01-C10/C11 local rebuild
P09	Optimization and constrained problems	p09_optimization.md	U05	U05-C07/C08 local derivations
P10	ML foundations and evaluation	p10_ml_foundations.md	U08	U08-C01 local protocol
P11	Neural networks and autodiff	p11_neural_nets.md	U03, U05	U03-C03, U05-C04 local mechanics
P13	Language and sequence modelling	p13_language.md	U07	U07-C03/C04 local rebuild
P14	Transformer mechanics	p14_transformer.md	U07	U07-C06 local mechanics
P15	Hardware and computer architecture	p15_hardware.md	U07	U07-C09 local cost model
P18	Bayesian inference, latent variables, sampling	p18_bayesian.md	U02, U03, U06	U02 is the local rebuild
P22	Experimental method and research literacy	p22_experiments.md	U08	U08-C07/C10 local protocol

Entry diagnostic

Answer closed-book. Each question names its remediation.

1. Compute $3/8 + 1/4$ as a decimal. (P01. Remediation: U01-C01 toy.)

2. Write $\log(a*b)$ and $\log(a/b)$ in terms of $\log a$ and $\log b$. (P01.)

3. A fair coin lands heads 3 times in 8 flips. Give the empirical probability of heads. (P06. Remediation: U01-C08.)

1. State Bayes rule in words, then in symbols. (P06/P18. Remediation: U02-C03.)

1. Define expectation of a discrete variable with 3 outcomes. (P06.)

2. A density integrates to 1.4 over its domain. What is wrong? (P08. Remediation: U01-C03.)

1. Compute KL divergence between Bernoulli(0.5) and Bernoulli(0.25) in nats. (P08. Remediation: U01-C11.)

1. Differentiate $f(x) = x^3 * \exp(x)$. (P05. Remediation: U04-C03.)

2. What is the determinant of $\text{diag}(2, 0.5)$? What does it say about area? (P04. Remediation: U04-C02.)

1. Write one step of gradient descent in symbols. (P09. Remediation: U05-C11.)

1. A model scores 0.9 accuracy on train and 0.6 on test. Name two possible causes. (P10. Remediation: U08-C01.)

1. State one requirement of a falsifiable hypothesis. (P22. Remediation: U08-C10.)

Pass bar: 9 of 12 with correct reasoning. Any fail routes to the named shared module first, then the unit-local remediation, then a retry of that question only.

Notation and shapes

Date: 2026-10-06. Baseline: October 6, 2026.

Every symbol below is defined before its first lesson use. Lessons cite this page instead of redefining symbols.

Probability

- x : one data point. X : the random variable. x_i : the i -th sample.
- $p(x)$: a probability density or mass function. $p_{\text{data}}(x)$: the true data law. $p_{\text{theta}}(x)$: the model law with parameters θ .
- $q(z|x)$: an approximate posterior (encoder). $p(z)$: a prior over latents. $p(x|z)$: a likelihood (decoder).
- $E_{x \sim p}[f(x)]$: expectation of f under p . E_n : empirical mean over n samples.
- $D_{\text{KL}}(p||q)$: KL divergence from p to q , in nats (natural log).
- $H(p)$: entropy of p . $H(p,q)$: cross-entropy.
- $N(\mu, \sigma^2)$: Gaussian with mean μ and variance σ^2 .
- $\text{supp}(p)$: support of p , the set where $p(x) > 0$.

Shapes

- D : data dimension. x in $\mathbb{R}^D$. d : latent dimension. z in $\mathbb{R}^d$.
- n : sample count. X in $\mathbb{R}^{n \times D}$: one row per sample.
- θ : parameter vector in $\mathbb{R}^P$. P : parameter count.
- W in $\mathbb{R}^{m \times n}$: matrix with m rows, n columns. b in $\mathbb{R}^m$: bias.
- J in $\mathbb{R}^{D \times D}$: Jacobian of a map $f: \mathbb{R}^D \rightarrow \mathbb{R}^D$.
- T : diffusion step count. t in $\{1..T\}$ or continuous t in $[0,1]$.
- K : codebook size. e_k in $\mathbb{R}^d$: the k -th code vector.
- V : vocabulary size for discrete tokens. L : sequence length.

Calculus and linear algebra

- $\text{nabla}_x f$: gradient of f with respect to x , shape of x .
- $\text{nabla}_x \log p(x)$: the score, shape of x .
- $\det J$ or $|\det J|$: Jacobian determinant, a scalar volume factor.
- $\text{diag}(v)$: diagonal matrix with v on the diagonal.
- $\|x\|_2$: Euclidean norm. $\langle x, y \rangle$: inner product.

Objectives

- $L(\theta)$: a loss (minimize). $J(\theta)$: an objective (context decides sign).
- ELBO: evidence lower bound. $\log p(x) \geq \text{ELBO}$.
- E_q : expectation under q . MC: Monte Carlo.

Code conventions

- numpy as `np`. `float64` for toy math unless stated.
- `rng = np.random.default_rng(seed)`. Seeds are printed with every figure.
- Shapes appear in comments as `# (n, D)`.

Units

- Probabilities are unitless in $[0,1]$. Densities carry $1/[x]$ units and can exceed 1. Nats for log-base- e information. Bits for log-base-2.
- Compute cost in FLOPs. Memory in bytes. Time in wall seconds on CPU unless stated.

Glossary

Date: 2026-10-06. Baseline: October 6, 2026.

- Amortization: one shared network predicts per-point variational parameters instead of optimizing each point separately.
- Base density: the simple law (often Gaussian) a flow transforms.
- Change of variables: the density transform under an invertible map, with the Jacobian determinant as volume correction.
- Codebook: the finite set of vectors a VQ model quantizes latents to.
- Coupling layer: a flow layer that transforms half the coordinates with parameters from the other half, giving a triangular Jacobian.
- DDIM: Denoising Diffusion Implicit Models, a deterministic sampler for diffusion models with fewer steps.
- DDPM: Denoising Diffusion Probabilistic Models, the standard discrete-time diffusion training and sampling recipe.
- Causal mask: a lower-triangular mask that stops each position from reading future positions.
- Chain rule: $p(x_1..x_L)$ as a product of conditionals, one per position.
- Classifier-free guidance: steering a diffusion sampler by extrapolating between conditional and unconditional scores.
- Density ratio: $p(x)/q(x)$, the object a discriminator estimates without modeling either density directly.
- Discriminator optimum: $D^*(x) = p(x)/(p(x)+q(x))$, the best

real-versus-fake classifier for a fixed generator.

- Denoising: training on noisy samples to learn the score of the noisy law. Tweedie turns the score into a clean estimate.
- ELBO: evidence lower bound. A tractable lower bound on log likelihood, tight exactly when the approximate posterior matches the true posterior.
- Empirical distribution: the law that puts mass $1/n$ on each observed sample.
- Exposure bias: the train-sample mismatch of teacher forcing. The model never learns to recover from its own errors.
- Falsification: stating what would prove a claim wrong before running the experiment.
- FID: Frechet Inception Distance. Compares Gaussian fits to feature means and covariances. Blind past second moments.
- Flow matching: training a continuous-time velocity field to match a target probability path, a labeled extension in U07-C11.
- Forward KL: $D_{KL}(p||q)$, mean-seeking, covers all of p . Reverse KL: $D_{KL}(q||p)$, mode-seeking.
- GAN: Generative Adversarial Network. A generator and a discriminator trained against each other.
- Gradient penalty: a soft 1-Lipschitz enforcement that penalizes the critic's gradient norm away from 1 on interpolated points.
- Guidance: conditioning a diffusion sampler toward a class or prompt, e.g. classifier-free guidance.

- Held-out likelihood: the likelihood on data the model did not train on. The honest generalization score.
- Identifiability: whether distinct parameters give distinct observable laws. Non-identifiable models admit silent parameter swaps.
- Implicit distribution: a law defined only by a sampling procedure, with no tractable density formula.
- Jacobian: the matrix of partial derivatives of a vector map.
- Jensen inequality: $E[\phi(X)] \geq \phi(E[X])$ for convex ϕ . The engine of the ELBO.
- Jensen-Shannon divergence: the symmetric divergence the GAN game minimizes at the discriminator optimum. Bounded by $\log 2$.
- KV cache: stored keys and values that make autoregressive generation $O(L^2)$ instead of $O(L^3)$.
- Langevin dynamics: gradient ascent on log-density plus noise, which samples from the density in the limit.
- Likelihood: $p(\text{data} \mid \text{parameters})$, the parameters' score on fixed data.
- Lipschitz constraint: a bound on how fast a function can change, used in WGAN to enforce the Kantorovich dual.
- Marginalization: summing or integrating out variables to get a marginal law.
- Mode collapse: a generator that covers a subset of the data modes.
- Mode covering: a fit that spreads mass over all data modes, often at the cost of blur.

- NLL: negative log-likelihood. The training loss for exact-likelihood models.
- Optimal transport: the cheapest way to move mass from one law to another. The 1-Wasserstein distance is its cost.
- Perplexity: $\exp(\text{NLL per token})$. The effective branching factor of a sequence model.
- Posterior: $p(z|x)$, the law over latents given data.
- Posterior collapse: a VAE whose approximate posterior equals the prior on some dimensions, so those latents carry no information.
- Precision (generative): fraction of model samples on the data manifold. Recall: fraction of data modes the model covers.
- Probability flow ODE: the deterministic twin of the reverse SDE with identical marginals. DDIM discretizes it.
- Rate/distortion: the tradeoff between latent bit cost and reconstruction error.
- Reparameterization: writing a sample as a deterministic function of parameters plus fixed noise, so gradients flow through sampling.
- Score: $\nabla_x \log p(x)$, the gradient of log-density.
- Score matching: fitting a model to the score instead of the density, which avoids the normalizing constant.
- Straight-through estimator: a biased gradient for discrete choices that copies the upstream gradient through the hard decision.
- Support: the set where a density is positive. Support mismatch

breaks likelihood and divergences.

- Teacher forcing: training an autoregressive model on ground-truth prefixes instead of its own predictions.
- Tweedie: $E[x_0 | x] = x + \sigma^2 \text{grad} \log p(x)$. The posterior mean from a noisy observation under Gaussian noise.
- VAE: Variational Autoencoder. An autoencoder trained with the ELBO.
- WGAN: Wasserstein GAN. Replaces the Jensen-Shannon objective with an optimal-transport distance.

Cheatsheet

Date: 2026-10-06. Baseline: October 6, 2026.

One dense page per unit family. Formulas use the notation fixed in notation_and_shapes.md. Section pointers name the lesson that proves each line. Honesty banner applies throughout.

U01, probability and density estimation

Formula / shape	Trap	Decision rule
Density integrates to 1, mass sums to 1	Reading $p(x)$ as a probability	Integrate before you believe
Likelihood $L(\theta) = \prod p(x_i, \theta)$	Calling it a function of the data	It moves with θ
Empirical law: mass $1/n$ per sample	Trusting $n = 12$	Watch it converge with n
Entropy $H = -E[\log p]$	Units: nats vs bits	State the base
$KL(p \parallel q)$	q directed	Symmetrizing it Direction is the penalty

U02, latent models and variational inference

Formula / shape	Trap	Decision rule
ELBO $= E_q[\log p(x, z) - \log q(z)]$	Calling it the likelihood	It is a floor
Gap $= KL(q \parallel p)$		posterior) Ignoring the gap Tighten q or label it
Forward KL covers, reverse seeks	Wrong direction for the job	Cover for recall, seek for sharpness
$z = \mu + \sigma \epsilon$	Detaching ϵ wrongly	ϵ carries no gradient anyway
Amortized q : one net for all x	The amortization gap	Compare against per-point q

U03, autoencoders and representation

Formula / shape	Trap	Decision rule
VAE loss = recon + KL	KL = 0 looks efficient	Zero KL is collapse
Gaussian KL closed form	Sampling it instead	Use the formula
beta-VAE: recon + beta x KL	beta too large kills latents	Ride the frontier, pick beta by it
VQ: nearest codebook vector	Straight-through bias 0.6011	Measure the bias on a toy
Rate-distortion frontier	More rate always helps	Only up to the frontier

U04, flows

Formula / shape	Trap	Decision rule
$p_x = \frac{\det J}{\det J^T}$ $p_z(f^{-1}(x)) \times \log p(y) = \log \det J - \log p_z - \log \det J^T$	Dropping \ The sign	$\det J$ Minus. Always minus
Triangular $\det = \text{diagonal product}$	A stray nonzero above diagonal	Audit the structure
MAF: density 1 pass, sample D steps Accumulate log-dets	Training a MAF to generate Linear-space products	Match the cheap direction to the job 100 layers of 1e6 overflow to inf

U05, adversarial and ratio estimation

Formula / shape	Trap	Decision rule
$D^*(x) = p/(p+q) = r/(1+r)$	$D^* = p/q$	The +q in the denominator
$\max_D V = -\log 4 + 2 \text{ JSD}$	Assuming D is optimal	It never is. The identity is theory
$\text{JSD} \leq \log 2$, saturates disjoint	100% accuracy as success	Saturation means zero gradient
W1 dual needs 1-Lipschitz f	Unconstrained critic	The value explodes
Gradient penalty, $\lambda = 10$	Penalty as proof	It constrains interpolations only
Mode collapse: recall 0	Precision alone	The pair or nothing

U06, scores and diffusion

Formula / shape	Trap	Decision rule
$s(x) = \text{grad} \log p(x)$	Gradient of the density	Log first, then gradient
$x_t = \sqrt{a_{t-1}} x_0 + \sqrt{1 - a_{t-1}} \epsilon$	Simulating the chain	The closed form is $O(1)$
DDPM loss = $E\ \epsilon - \epsilon_{\hat{t}} \ ^2$		Calling it a likelihood The reweighting breaks the bound
Tweedie: $x + \sigma^2 \text{ score}$	Using it as a sampler	It returns the mean
DDIM $\eta = 0$: deterministic	Striding for free	Validate quality vs steps
Guidance: $s + w(s_c - s_u)$	w too large	Watch for overshoot artifacts
Boundary score blows up	ODE likelihood without a floor	Name $\sigma_{\min}$ or void the number

U07, autoregressive models

Formula / shape	Trap	Decision rule
$p(x) = \text{product } p(x_{:t})$	$x_{:t}$	Forgetting the conditioning Each factor sees the past
Perplexity = $\exp(\text{NLL}/L)$	Across tokenizations	Bits per character first
Causal mask: lower triangular	One unmasked layer	The future leaks everywhere
Attention: $\text{soft-max}(QK^T/\sqrt{d})V$	Dropping the scale	Softmax saturates at init
Cost $O(L^2 d)$, cache 2Ld bytes	Generating without cache	$O(L^3)$: throughput dies
Temperature T on logits	High T without truncation	Truncate, then temper
Flow matching: regress $x_1 - x_0$	Crossing paths	OT couplings straighten them

U08, evaluation

Formula / shape	Trap	Decision rule
Held-out loglik	Tuning on it	Lock it away
Bound vs exact	Ranking across the gap	Exact vs exact only
Precision / recall	One without the other	The pair is the minimum
FID: moments only	FID 0 as verdict	Build the bimodal counterexample
NN memorization test	Eyeballing samples	Histogram the distances
Compute-matched budgets	The idea vs the money	Fix params, flops, inference cost
Uncertainty over seeds	Single-seed leaderboard	5 seeds minimum, report std
Pre-registered failure rule	HARKing	Write it before the run

Cross-unit decision rules

Choice	Rule
Need exact likelihood	Flow or AR. Never a GAN, never a bare VAE number
Need fast sampling	GAN, DDIM, or IAF. Not MAF, not DDPM-1000
Laws start far apart	Wasserstein-flavored. Not JSD
Discrete data	AR or discrete diffusion. Not Gaussian diffusion raw
Compare two models	Matched budgets, frozen metrics, seed error bars
Report a number	Name the assumption it needs

Crash course

Date: 2026-10-06. Baseline: October 6, 2026.

Eight units, about five minutes each. Every worked number below comes from the named lesson section. The crash course points, the lessons prove. Honesty banner from README.md applies: this is a provisional independent bridge, not asserted official syllabus.

U01, Probability, measures, and density estimation (1 lessons/u01)

Probability masses sit on points and sum to 1. Densities sit under curves and integrate to 1. The empirical distribution puts mass $1/n$ on each sample and approaches the truth as n grows. Likelihood scores fixed data under moving parameters. Entropy is expected surprise. KL divergence is directed: $KL(p||q)$ punishes q for missing p 's mass.

Worked number (U01): 12 draws from (0.5, 0.333, 0.167). Loglik under the true law -12.137 versus -13.183 under a wrong law. The data pick the truth by 1.046 nats.

Memory aid: mass sums, density integrates. The likelihood is a function of the parameters, not the data.

Self-test: Q: Why is $p(x) = 0.3989$ at the mode not a probability? A: A density value is not a mass. Only integrals of it are probabilities.

U02, Latent models and variational inference (1 lessons/u02)

The marginal $p(x) = \sum_z p(x, z)$ is intractable when z is rich. Bayes inverts to the posterior. Jensen on the concave log gives the ELBO: $\log p(x) \geq E_q[\log p(x, z) - \log q(z)]$. The gap is $KL(q||\text{posterior})$: zero exactly when q matches the posterior. Forward KL covers, reverse KL seeks. The reparameterization trick moves noise outside the parameters so gradients flow.

Worked number (U02): $\log p(R) = -0.5108$, ELBO -0.5390, gap 0.0282. The bound sits below the truth by exactly the KL.

Memory aid: the ELBO is a floor. The gap is the distance from your q to the truth.

Self-test: Q: When is the ELBO tight? A: When q equals the posterior. Then the gap KL is zero.

U03, Autoencoders and representation objectives (lessons/u03)

A deterministic autoencoder compresses and rebuilds. A VAE trains the ELBO: reconstruction minus the KL to the prior. The Gaussian KL has a closed form. Posterior collapse kills the latents: the decoder ignores z . Beta scales the rate price on the rate-distortion frontier. VQ replaces the continuous latent with a codebook and pays the straight-through bias.

Worked number (U03): recon -2.1516, KL 0.9013, ELBO -3.0529, 2.2022 bits/dim. The bound, labeled as a bound.

Memory aid: rate buys distortion. Dead latents have zero KL.

Self-test: Q: KL reads 0.0003 per dim but reconstructions are sharp. What happened? A: Collapse. The decoder memorized and ignores z .

U04, Change of variables and normalizing flows (lessons/u04)

An invertible map plus the Jacobian determinant gives exact density: $p_x(x) = p_z(f^{-1}(x)) \times |\det J|$. Coupling layers make the Jacobian triangular, so the determinant is the diagonal product at $O(D)$ cost. Forward samples, inverse scores. MAF is cheap at density eval, IAF at sampling. One affine layer of a Gaussian stays Gaussian. Accumulate log-determinants or overflow.

Worked number (U04): $\log p(y) = -2.4629 - 0.2231 = -2.6860$ nats at the toy point. Exact, not a bound.

Memory aid: the minus sign. Forgetting it reports -2.2398.

Self-test: Q: Why triangular? A: The determinant becomes the diagonal product. $O(D^3)$ becomes $O(D)$.

U05, Adversarial and ratio-based estimation (lessons/u05)

An implicit law samples without a density. The discriminator estimates the density ratio through $D^* = p/(p+q)$. At the optimum the game minimizes the Jensen-Shannon divergence: $\max_D V = -\log 4 + 2 \text{JSD}$. JSD saturates on disjoint laws and the gradient dies. The Wasserstein dual keeps a linear signal but needs the 1-Lipschitz constraint. The gradient penalty enforces it only on interpolations: a heuristic, not a proof. Mode collapse is a recall failure the loss cannot see.

Worked number (U05): $D^*(0) = 0.6225$, $\text{JSD} = 0.1114$, $V^* = -1.1635$. Disjoint: $\text{JSD} = 0.6931 = \log 2$, $W1 = 5.0$.

Memory aid: the critic is a ratio meter. Perfect accuracy means saturation, not convergence.

Self-test: Q: The discriminator hits 100 percent. Good news? A: No. JSD is saturated at $\log 2$ and the generator gets zero gradient.

U06, Scores, diffusion, and stochastic dynamics (lessons/u06)

The score is $\text{grad log } p$: it points uphill with no normalizer. Score matching fits it without the trace via denoising. Tweedie turns a noisy score into a clean estimate. The DDPM chain destroys signal on a schedule. The loss predicts the noise. DDIM strides the chain deterministically. Langevin adds calibrated noise to uphill steps. Guidance steers the score. Solvers trade steps for accuracy. Boundary scores blow up: likelihoods need a named noise floor.

Worked number (U06): Tweedie 1.2 $\rightarrow$ 0.0 exactly. DDIM step 1.70 $\rightarrow$ 1.7630. DDPM loss 0.005 on the toy batch.

Memory aid: the score needs no partition function. The noise schedule is the curriculum.

Self-test: Q: Why does denoising avoid the trace term? A: The model is never differentiated. Tweedie links noise prediction to the score directly.

U07, Autoregressive models and modern bridges (lessons/u07)

The chain rule factors the joint exactly: order is the model. Teacher forcing trains on true prefixes. Sampling sees its own outputs (exposure bias). The causal mask is the arrow of time. Attention is a weighted past, costed at $O(L^2 d)$ with a KV cache. Temperature reshapes the draw without retraining. Masked models have no valid joint: pseudo-perplexity sums to 1.0639, not 1.0. Flow matching regresses the velocity of a chosen path directly.

Worked number (U07): $\log p(\text{abc}) = -2.9957$, perplexity 2.7144. Attention head row 2: weights (0.3302, 0.6698), output (3.3395, 4.3395).

Memory aid: exact likelihood, serial sampling. Compare perplexity only within one tokenization.

Self-test: Q: Why is the AR likelihood exact? A: The chain rule is an identity. No bound, no sampling.

U08, Evaluation, inference, and experimental synthesis (lessons/u08)

Held-out likelihood is the honest score. Bounds are floors: never rank a bound against an exact number silently. Precision is quality, recall is coverage: report both. FID sees only moments: FID 0 with the wrong shape is the counterexample. Memorization hides from likelihood: run the NN test. Match compute before comparing ideas. Synthetic truth is the laboratory. Report seed uncertainty. Pre-register the failure criterion: falsify, do not confirm.

Worked number (U08): held-out A -10.1073 versus B -70.949. Precision 0.4708/0.9555, recall 1.0/0.5. NN 0.0066 versus 0.4666.

Memory aid: one metric is one blind spot with confidence.

Self-test: Q: FID is 0. Done? A: No. Match the moments and miss the shape: FID 0, truth bimodal, model unimodal.

The whole course in eight lines

U01: mass sums, density integrates, likelihood scores. U02: the ELBO floors the marginal, the gap is a KL. U03: rate buys distortion, dead latents cost nothing. U04: invert, correct the volume, read the exact number. U05: the game minimizes JSD at the optimum, which never holds. Watch the geometry. U06: learn the uphill direction at every noise scale, then walk back. U07: order the factors, pay the quadratic context, sample serially. U08: no single number is honest. Falsify.